Pure fibres of the four-site matching map

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Abstract

The four-site matching map sends six pair tensors on four three-dimensional complex vector spaces to the sum of their products over the three perfect matchings. Over a general output its fibre is known to be a single orbit of a three-dimensional torus. We study the fibre over a pure tensor, that is, a nonzero product of four vectors. We give complete parametrizations of this fibre, organized by nine support and rank patterns after quotienting each local space by its target line. The fibre has dimension 19 and exactly twelve irreducible components of that dimension, each isomorphic to $\mathbb{A}^{18} \times \mathbb{G}_m$. On the open subset where all six pair tensors are nonzero, the dimension is 15, again with twelve components of maximal dimension. The corresponding space of three ordered nonzero summands, one decomposable across each of the three pair partitions, has dimension 12. The proof combines a projected-support obstruction, elementary tensor syzygies, and an incidence calculation for isotropic planes. The parametrizations allow zero blocks and drops of whole rank; lower-dimensional components are not classified.

1 The fibre and the main results

Let $V_1, \dots, V_4$ be three-dimensional complex vector spaces. Put $U_{ij} = V_i \otimes V_j$ for $i < j$, and consider the polynomial map

$$\Phi : \prod_{i<j} U_{ij} \longrightarrow V_1 \otimes V_2 \otimes V_3 \otimes V_4, \qquad \Phi(W) = W_{12}W_{34} + W_{13}W_{24} + W_{14}W_{23}.$$

A product in this formula is the tensor product on disjoint sites, restored to the order $1, 2, 3, 4$, with no sign. Thus the three terms correspond to the three perfect matchings of the complete graph on four labelled vertices. The map Φ is the matching tensor of that graph in the sense of [7], here with all local dimensions equal to three. Matching tensors arise from the description of optical quantum experiments by graphs [10]; see the introduction of [7]. No question from that subject is decided here. For three colours and no internal vertices, the same map gives the full-sector coefficients in the model of [9, Section 2], of which no result is used here.

The closure of the image of Φ is the join of the three cones of tensors decomposable across $12|34$, $13|24$ and $14|23$ [7, Proposition 7.1 and its proof], and over a general point of that closure the whole fibre is a single orbit of the three-dimensional torus of vertex rescalings [7, Theorem 4.5]. That paper lists a description of the exceptional outputs among the things it does not prove. Here an output is called *exceptional* if its fibre is not a single such orbit. Call a tensor in $V_1 \otimes V_2 \otimes V_3 \otimes V_4$ *pure* if it is a nonzero product of four vectors, one from each V_i, and call the fibre of Φ over a pure tensor a *pure fibre*. A pure tensor lies in all three cones and is exceptional: its fibre has dimension 19, not three (Theorem 1.1).

Fix nonzero $c_i \in V_i$ and $\tau \in \mathbb{C}^*$. We study the reduced affine algebraic set

$$F_\tau = \{W : \Phi(W) = \tau c_1 c_2 c_3 c_4\}.$$

No quotient by changes of basis is taken. Let

$$F_\tau^{\mathrm{nz}} = \{W \in F_\tau : W_{ij} \neq 0 \text{ for all } i < j\}.$$

This is an open subvariety. Its defining nonvanishing condition is equivalent to requiring all three matching products to be nonzero.

The same question can be asked of the three products alone: in which ways is a pure tensor a sum of three nonzero tensors, one decomposable across each of the three pair partitions? Let $\mathcal{S}^\times_{ij|kl}$ be the set of nonzero tensors of the form XY with $X \in U_{ij}$ and $Y \in U_{kl}$, the product being formed as in the definition of Φ. Define

$$D_\tau = \left\{(A,B,C) \in \mathcal{S}^\times_{12|34} \times \mathcal{S}^\times_{13|24} \times \mathcal{S}^\times_{14|23} : A + B + C = \tau c_1 c_2 c_3 c_4\right\}.$$

The summands are ordered and their three partitions are prescribed. They need not be pure.

Theorem 1.1. *With the preceding hypotheses:*

1. $\dim F_\tau = 19$. *Exactly twelve irreducible components have dimension* 19*, and each is isomorphic to* $\mathbb{A}^{18} \times \mathbb{G}_m$.
2. $\dim F_\tau^{\mathrm{nz}} = 15$. *Exactly twelve irreducible components of* F_τ^{nz} *have dimension* 15.
3. *The matching-product map* $F_\tau^{\mathrm{nz}} \to D_\tau$ *is a Zariski locally trivial bundle with fibre* $(\mathbb{G}_m)^3$. *Consequently* $\dim D_\tau = 12$*, and exactly twelve irreducible components of* D_τ *have dimension* 12.

These assertions specify the components of maximal dimension, not the complete irreducible decompositions of these varieties.

The large components of the full fibre are easy to exhibit. Choose a vertex i, two of its neighbours j, k, and write z for the remaining vertex. Set

$$W_{jz} = W_{kz} = 0, \qquad W_{iz} = q c_i c_z, \qquad W_{jk} = (\tau/q) c_j c_k, \qquad q \neq 0, \tag{1}$$

and leave W_{ij}, W_{ik} arbitrary. The two free blocks do not contribute to the output because their complementary blocks vanish. The first assertion of the theorem includes the global upper bound and the fact that these are all components of that dimension. The second and third assertions concern cancellation among three nonzero products, where no block is free in this way. There the components of dimension fifteen are the closures in F_τ^{nz} of the strata whose projected support, defined below, is a triangle with ranks $(2,1,1)$; the chart (20) parametrizes these strata, and Section 6 gives an explicit point.

Here is the classification used to prove the theorem. Let $\pi_i : V_i \to V_i/\mathbb{C}c_i$, and write $\overline{W}_{ij} = (\pi_i \otimes \pi_j) W_{ij}$. The *projected support graph* has edge ij precisely when $\overline{W}_{ij} \neq 0$. A zero projected block may have a nonzero whole block.

A block W_{ij} is *pure* if it is a nonzero multiple of $c_i c_j$. The *star* at a vertex is the set of the three edges at it, and that vertex is its *centre*. A *wedge* is a projected support consisting of two adjacent edges, *centred* at their common vertex. The other two vertices of a wedge, and the three vertices other than the centre of a star, are its *leaves*. If the projected support is a triangle, the blocks joining its vertices to the fourth vertex are its *spokes*; if its projected ranks are $(2,1,1)$, the edge of rank two is the *high* edge and the other two are the *low* edges. A point of F_τ is *active* if it lies in F_τ^{nz}; an active stratum is the intersection of a stratum with F_τ^{nz}, and an active component is an irreducible component of F_τ^{nz}. The exceptional site in Lemma 3.2 only labels a row of its table; it is unrelated to exceptional outputs.

Theorem 1.2. *Every point of F_τ, up to simultaneous relabelling of the four sites, has exactly one of the nine projected support and rank patterns in Table 1. Every pattern occurs. Section 3 gives complete whole-tensor parametrizations of all nine types. The dimension in the table is that of a stratum with its labelled support and rank assignment fixed.*

Projected support	Nonzero projected ranks	Dimension	Nonzero-block part
Empty	—	11	≤ 11
One edge	1	14	12
One edge	2	15	13
Adjacent edges	1, 1	17	13
Adjacent edges	2, 1	18	empty
Adjacent edges	2, 2	19	empty
Triangle	1, 1, 1	14	14
Triangle	2, 1, 1	15	15
Four-cycle	1, 1, 1, 1	12	12

Table 1: Complete projected types. The last column concerns the intersection with F_τ^{nz}; only an upper bound is needed in its first row.

The classification is a parametrization by finitely many types, with redundant parameters where explicitly indicated. It is not a classification of orbits under local linear groups. In particular the empty-support parametrization uses isotropic subspaces, rather than a finite list of coordinate representatives.

Relation to earlier work

The problem is a nonminimal decomposition problem at an exceptional output, for three different paired Segre cones. Generic uniqueness results for joins, such as Ballico's Proposition 8 [1], assume a general point of the relevant join and its prescribed join rank. The target here belongs to each constituent Segre cone. Nonminimal decomposition results of Lovitz and Petrov [4, Section 6.2] concern product tensors relative to one fixed factorization; the three factorizations above differ. Neither statement determines this fibre.

The ingredients of our proof have familiar counterparts. The intersection of two crossing partition Segre varieties is the Segre variety for their common refinement; see the proof of Theorem 6.1 in Section 6 of Gesmundo, Han and Lovitz [2], in the numbering of arXiv:2407.16767v2. This accounts for the rank-one factors in the four-cycle case. Zero-slice decompositions, as studied by Karam [3] (Proposition 3.2 in the numbering of arXiv:2308.07101v1), provide a broader setting for the elementary syzygies used after a triangle reduction. The two-term case used repeatedly below, in which a relation $xA+yB=0$ between nonzero tensors A, B, with nonzero vectors x, y at two different sites, forces $A = hy$ and $B = -hx$ for a common factor h, is the two-leaf case of [8, Theorem 2.1]; see also Corollary 9.2 there. The proof of Lemma 2.1 contracts with a covector chosen outside finitely many kernels. The same generic-contraction and finite-union device appears, in another setting, in Section 3 of the public provisional note [5]; no proof here depends on that note. The quadratic-form calculations below use ordinary isotropic-subspace geometry [6]. The surjectivity of the bundle in Theorem 1.1(3) is the four-vertex construction in the proof of [7, Proposition 7.1].

Our subject is the complete fibre over a pure target in this fixed format, including its whole-tensor lifts and its sharp dimension bounds. The result does not assert a general theorem on

partition rank [11] or an exclusion for larger matching systems. Bounded searches by AI agents found no earlier statement of the results of this paper; this is not proof that they are new.

2 Exhaustion of the projected supports

All tensor products in the proofs retain the actual given blocks. We may choose splittings $V_i = \mathbb{C}c_i \oplus H_i$, with $\dim H_i = 2$, when coordinates are useful.

Lemma 2.1. *The projected support graph of a point of F_τ has maximum degree at most two.*

Proof. Suppose that all three projections incident to a vertex i are nonzero. Their contractions are three nonzero linear maps from $\mathrm{Ann}(c_i)$ to the corresponding quotient spaces. A finite union of proper linear subspaces cannot cover a complex vector space. Choose $\alpha \in \mathrm{Ann}(c_i)$ outside their three kernels.

Write the other vertices, the leaves of the star at i, as j, k, z. The whole contractions f_j, f_k, f_z of the incident blocks by α have nonzero quotient images, so each f_r is independent of c_r. Choose fixed dual functionals δ_r, ϵ_r satisfying

$$\delta_r(c_r) = 1, \quad \delta_r(f_r) = 0, \qquad \epsilon_r(c_r) = 0, \quad \epsilon_r(f_r) = 1.$$

Contract the leaf slots by $\delta_j + u\epsilon_j, \delta_k + v\epsilon_k, \delta_z + w\epsilon_z$. Keeping the centre slot whole gives an identity

$$A(u)\phi(v, w) + B(v)\psi(u, w) + C(w)\chi(u, v) = \tau c_i.$$

The scalar factors have the displayed absent-variable dependencies because their blocks do not meet the missing leaf. Applying α yields

$$u\phi(v, w) + v\psi(u, w) + w\chi(u, v) = 0.$$

Evaluation at $(1, 0, 0), (0, 1, 0), (0, 0, 1)$ gives $\phi(0, 0) = \psi(0, 0) = \chi(0, 0) = 0$. The whole identity at the origin would then say $0 = \tau c_i$, a contradiction. This proof allows zero leaf blocks and arbitrary components along the target lines. □

Lemma 2.2. *If the projected support is a triangle, at least two of its edges have rank one. More precisely, if z is the isolated projected vertex, there is a unique triangle vertex i whose whole spoke to z is a nonzero multiple of $c_i c_z$. The two triangle edges incident to i have rank-one projections.*

Proof. Write the triangle vertices as l, m, t, with

$$L = W_{lm}, \quad P = W_{lt}, \quad R = W_{mt}, \qquad Q = W_{lz}, \quad S = W_{mz}, \quad K = W_{tz}.$$

The three triangle projections are nonzero; the other three vanish. Projecting at l, m, z in the full equation isolates $\overline{L}(\mathrm{id}_t \otimes \pi_z)K = 0$. Thus $K = p_t c_z$. The other two triple projections give $Q = p_l c_z, S = p_m c_z$. Cancelling the fixed nonzero factor c_z,

$$Lp_t + Pp_m + Rp_l = \tau c_l c_m c_t.$$

Quotienting each triangle space by its actual line $\mathrm{span}(p_r)$, with $\mathrm{span}(0) = 0$, kills the left side. Some c_i must therefore belong to $\mathrm{span}(p_i)$. Write $p_i = \lambda c_i$, where $\lambda \neq 0$.

Let j, k be the other triangle vertices and put $A = W_{ij}, B = W_{ik}, C = W_{jk}$. Then

$$Ap_k + Bp_j + \lambda c_i C = \tau c_i c_j c_k.$$

If $\bar{p}_j = 0$, projection at i, j forces $p_k = 0$, and projection at i, k then forces $p_j = 0$. The equation would imply $\bar{C} = 0$, a contradiction. Similarly $\bar{p}_k \neq 0$. This also proves uniqueness of the pure spoke. Indeed a pure spoke at j or k would mean $p_j \in \mathbb{C}c_j$ or $p_k \in \mathbb{C}c_k$, that is, $\bar{p}_j = 0$ or $\bar{p}_k = 0$.

Project only at i, writing A', B' for the resulting tensors. The identity $A'p_k + B'p_j = 0$, with $p_j, p_k \neq 0$, implies $A' = hp_j, B' = -hp_k$ for one $h \neq 0$: apply the quotient by $\operatorname{span}(p_j)$ at j, then cancel the nonzero factors. Choose a whole lift u of h. The complete whole blocks have the form

$$A = up_j + c_i v, \qquad B = -up_k + c_i w, \qquad C = \lambda^{-1}(\tau c_j c_k - vp_k - p_j w). \tag{2}$$

In particular $\bar{A} = h\bar{p}_j$ and $\bar{B} = -h\bar{p}_k$ have rank one. No initial rank-one assumption was used. □

Proof of the support assertion in Theorem 1.2. A simple graph on four vertices of maximum degree two is empty, one edge, two adjacent edges, two disjoint edges, a three-edge path, a triangle, or a four-cycle. The disjoint-edge and three-edge-path graphs each have exactly one supported perfect matching. Fourfold projection of $\Phi(W)$ would then be one nonzero product, whereas the projected target is zero.

For a four-cycle, take its missing edges to be lm, tz. The fourfold equation reads $\overline{P}\,\overline{S} = -\overline{Q}\,\overline{R} \neq 0$. Across $lt|mz$ the left side has rank one and the right side has rank $\operatorname{rank}\overline{Q}\,\operatorname{rank}\overline{R}$, since its matrix is a reordered Kronecker product. Both ranks on the right are one. The other crossing cut gives the same conclusion for P, S. Lemma 2.2 leaves only ranks 111 and 211 for a triangle. All remaining nonzero ranks are one or two because the local quotients have dimension two. This gives the nine types. The complete lift charts below establish sufficiency and nonemptiness. For instance, in (1), free blocks W_{ij}, W_{ik} of projected ranks $(0,0)$, $(1,0)$, $(2,0)$, $(1,1)$, $(2,1)$ and $(2,2)$ give the first six types of Table 1 in order; (13) with $v = q$ and $w = 0$ gives a triangle of ranks $(1,1,1)$; the point of Section 6 is a triangle of ranks $(2,1,1)$; and (15) with $\gamma = 1$ and all $u_i \in H_i \setminus \{0\}$ gives a four-cycle. □

3 Whole-tensor parametrizations

We give the whole tensors in each of the nine types. In this section (l, m, t, z) is an ordering of the four sites, and

$$L = W_{lm}, \quad P = W_{lt}, \quad Q = W_{lz}, \quad R = W_{mt}, \quad S = W_{mz}, \quad K = W_{tz}.$$

Juxtaposition denotes tensor product, restored without a sign to the indicated site order. The equation to be solved is

$$LK + PS + QR = \tau c_l c_m c_t c_z, \qquad \tau \neq 0. \tag{3}$$

Write $V_i = \mathbb{C}c_i \oplus H_i$, using the fixed complements to identify quotients with H_i. A bar on a vector is its quotient image; on an edge tensor it means projection at both endpoints. In particular,

$$\overline{W}_{ij} = 0 \quad \Longleftrightarrow \quad W_{ij} \in c_i \otimes V_j + V_i \otimes c_j. \tag{4}$$

A tensor described as free with prescribed projected rank may have arbitrary entries involving a c factor, including entries making its whole matrix singular. All scalar and vector parameters below are complex.

3.1 Empty projected support

Let $E = \mathbb{C}^6$, with coordinates indexed by the six unordered pairs of sites, and put

$$q(x) = x_{12}x_{34} + x_{13}x_{24} + x_{14}x_{23}, \qquad b(x,y) = q(x+y) - q(x) - q(y).$$

Let $S_i \subset E$ be the coordinate subspace on the three edges incident to i.

Proposition 3.1. *The solutions with all six double projections zero are parametrized as follows. Choose $a \in E$ with $q(a) = \tau$, a totally isotropic subspace $U \subset a^\perp$ for b, and linear maps*

$$B_i : H_i^* \longrightarrow U \cap S_i \qquad (1 \le i \le 4).$$

Necessarily $\dim U \le 2$. Define $u_{ij} \in H_i$ by $(B_i(x))_{ij} = x(u_{ij})$, and set

$$W_{ij} = a_{ij}c_ic_j + u_{ij}c_j + c_iu_{ji}. \tag{5}$$

These formulas give every solution of this type. The data a, B_i are unique; the choice of U becomes unique if $U = \sum_i \operatorname{im} B_i$ is required.

Proof. By (4), every such tuple has the unique form (5). Evaluate the tensor at covectors taking c_i to 1, and denote their restrictions to H_i by x_i. The six edge evaluations form the vector $a + \sum_i B_i(x_i)$. Thus the target identity is equivalent to

$$q\left(a + \sum_i B_i(x_i)\right) = \tau \quad \text{for all } x_i \in H_i^*.$$

This equivalence loses no coefficients: the expression is multiaffine in the four groups of variables, and homogenizing each group recovers the original four-tensor. Comparison of constant, linear and quadratic terms, followed by polarization, gives precisely

$$q(a) = \tau, \qquad b(a,U) = 0, \qquad b(U,U) = 0, \qquad U = \sum_i \operatorname{im} B_i.$$

The quadratic terms give $b(\operatorname{im} B_i, \operatorname{im} B_j) = 0$ for $i \ne j$; for $i = j$ this holds because each S_i is totally isotropic, no two edges at i being complementary. Conversely these conditions make the displayed polynomial constant. The form b is nondegenerate, since its matrix has three invertible complementary-edge blocks. As $b(a,a) = 2\tau \ne 0$, the space $a^\perp$ is nondegenerate of dimension five. For an isotropic U in that space, $U \subset U^\perp$, so $2\dim U \le 5$. □

3.2 A single projected edge

We first record the crossing decomposition needed at a zero opposite edge.

Lemma 3.2. *Suppose $\overline{P} = \overline{Q} = \overline{R} = \overline{S} = 0$ and $PS + QR = \tau c_l c_m c_t c_z$. The following list is exhaustive.*

1. *If $PS = 0$, then for some $a \ne 0$, $Q = ac_lc_z$, $R = (\tau/a)c_mc_t$, and $P = 0$ or $S = 0$. The unused partner is any zero-double-projection tensor.*
2. *If $QR = 0$, then for some $a \ne 0$, $P = ac_lc_t$, $S = (\tau/a)c_mc_z$, and $Q = 0$ or $R = 0$. The unused partner is any zero-double-projection tensor.*

3. *If both products are nonzero, all four edges have whole rank one. Choose one exceptional site and one row of the following table:*

	P	S	Q	R	*condition*
l	uc_t	sc_mc_z	vc_z	rc_mc_t	$su + rv = \tau c_l$
m	pc_lc_t	uc_z	qc_lc_z	vc_t	$pu + qv = \tau c_m$
t	c_lu	sc_mc_z	qc_lc_z	c_mv	$su + qv = \tau c_t$
z	pc_lc_t	c_mu	c_lv	rc_mc_t	$pu + rv = \tau c_z$

All named scalar parameters and both vectors u, v are nonzero. In each row the displayed equation explicitly determines v from u and the two scalars; the only additional restriction is $v \neq 0$. The choices of exceptional site may overlap.

Every listed choice is a solution.

Proof. If one product vanishes, the other equals the nonzero pure tensor. Factorization across its own edge bipartition gives the first two cases; a disjoint tensor product vanishes exactly when a whole factor vanishes.

Suppose both products are nonzero. Every edge has rank at most two by (4). If P had rank two, flatten $PS = \tau c_lc_mc_tc_z - QR$ across $(l, z) \mid (m, t)$. Its rank is $\operatorname{rank}(P)\operatorname{rank}(S)$, whereas the right side has rank at most two. Thus $S = s_ms_z$ has rank one and the flattening has rank two. Its left and right spaces are respectively

$$\operatorname{im}_l(P) \otimes \mathbb{C}s_z = \operatorname{span}(c_lc_z, Q), \qquad \mathbb{C}s_m \otimes \operatorname{im}_t(P) = \operatorname{span}(c_mc_t, R).$$

It follows that $S = sc_mc_z$, $Q = qc_z$, and $R = c_mv$, with $s \neq 0$. Hence $sP = \tau c_lc_t - qv$. Rank two of P forces $q \notin \mathbb{C}c_l$ and $v \notin \mathbb{C}c_t$, contradicting $\overline{P} = 0$. The same argument, with the other crossing cut, applies to each of Q, R, S. Thus all four have rank one.

Now PS and QR are nonzero simple four-tensors whose sum is nonzero and simple. Their factor lines can differ at at most one site: otherwise flattening at one differing site gives two independent factors on each side and therefore rank two. At every other site the common line must be the target line $\mathbb{C}c_i$. Absorbing the scalar factors gives exactly the four rows. Substitution proves their converses, including the case in which all four factor lines agree. □

Proposition 3.3. *Fix $r \in \{1, 2\}$. Suppose only $\overline{L}$ is nonzero and its rank is r. Then $K = \kappa c_tc_z$, and all solutions are the following.*

1. *If $\kappa = 0$, take L free with projected rank r, and take P, Q, R, S from Lemma 3.2.*

2. *If $\kappa \neq 0$ and both crossing products are nonzero, choose nonzero $p, q \in V_l$ and $r_0, s \in V_m$, and put*

$$\begin{gathered} P = pc_t, \quad Q = qc_z, \quad R = r_0c_t, \quad S = sc_z, \\ L = \kappa^{-1}(\tau c_lc_m - ps - qr_0). \end{gathered} \tag{6}$$

Require the matrix $M = \overline{p}\,\overline{s} + \overline{q}\,\overline{r_0}$ to have rank r. Explicitly, in any quotient bases, this means

$$\begin{cases} M \neq 0, \quad \det[\overline{p}, \overline{q}] \det[\overline{s}, \overline{r_0}] = 0, & r = 1, \\ \det[\overline{p}, \overline{q}] \det[\overline{s}, \overline{r_0}] \neq 0, & r = 2. \end{cases}$$

3. *If $\kappa \neq 0$ and $QR = 0$, necessarily $r = 1$. Choose $p \in V_l, s \in V_m$ with nonzero quotients, and set*

$$P = pc_t, \quad S = sc_z, \quad L = \kappa^{-1}(\tau c_lc_m - ps).$$

Choose any zero-double-projection Q, R with $Q = 0$ or $R = 0$.

4. *If $\kappa \neq 0$ and $PS = 0$, necessarily $r = 1$. Choose $q \in V_l, r_0 \in V_m$ with nonzero quotients, and set*

$$Q = qc_z, \quad R = r_0 c_t, \quad L = \kappa^{-1}(\tau c_l c_m - q r_0).$$

Choose any zero-double-projection P, S with $P = 0$ or $S = 0$.

The branches exhaust all zero and nonzero crossing products.

Proof. Project (3) at l, m, t, and then at l, m, z. The crossing terms vanish in each case, so the nonzero $\overline{L}$ forces both partial quotients of K to vanish. Hence $K = \kappa c_t c_z$. Write the other edges uniquely as

$$P = pc_t + c_l a, \quad Q = qc_z + c_l b, \quad R = r_0 c_t + c_m d, \quad S = sc_z + c_m e,$$

where $a, d \in H_t$, $b, e \in H_z$, and the other vectors are whole. Separating the four (t, z) sectors gives the equivalent full system

$$\begin{aligned} &\kappa L + ps + qr_0 = \tau c_l c_m, \\ &c_l sa + qc_m d = 0, \qquad pc_m e + c_l r_0 b = 0, \quad ae + db = 0. \end{aligned} \tag{7}$$

When $\kappa = 0$, L drops out and Lemma 3.2 applies.

Let $\kappa \neq 0$. The double projection of the first equation shows that at least one of $\overline{p}\,\overline{s}$ and $\overline{q}\,\overline{r_0}$ is nonzero. If the first is nonzero, projecting the second equation at m gives $a = 0$, and projecting the third at l gives $e = 0$. The remaining conditions are $qd = 0$, $r_0 b = 0$, and $db = 0$. If $Q, R \neq 0$, these force $b = d = 0$: for example $d \neq 0$ would imply $q = b = 0$, hence $Q = 0$. If $QR = 0$, the unused partner is unrestricted within (4). Interchanging the two crossing products proves the corresponding assertion when the second projected summand is nonzero. Both whole products cannot vanish, since that would make $\overline{L} = 0$. This proves all the stated forms. The determinant formula follows by writing $M = [\overline{p}, \overline{q}][\overline{s}, \overline{r_0}]^T$. A single nonzero projected outer product has rank one, explaining the last two branches. Finally the displayed formulas solve the entire system (7), which proves every converse. □

3.3 Two adjacent projected edges

Here $\overline{L}, \overline{P} \neq 0$ and $\overline{Q} = \overline{R} = \overline{S} = \overline{K} = 0$.

Lemma 3.4. *Every solution in this support has*

$$Q = qc_z, \quad S = sc_z, \quad K = kc_z, \qquad Lk + Ps + qR = \tau c_l c_m c_t, \tag{8}$$

where $q \neq 0$ and $R \neq 0$. Either both $\overline{s}, \overline{k}$ vanish or both are nonzero. If both vanish, write $k = \kappa c_t$, $s = \sigma c_m$, and use the unique decompositions

$$\begin{aligned} R &= r_H c_t + c_m t_H + \rho c_m c_t, \\ L &= L_H + x c_m, \qquad P = P_H + y c_t, \end{aligned}$$

with $r_H \in H_m$, $t_H \in H_t$, $L_H \in V_l \otimes H_m$, $P_H \in V_l \otimes H_t$. Then the reduced equation is equivalent to

$$\kappa L_H + q r_H = 0, \qquad \sigma P_H + q t_H = 0, \qquad \kappa x + \sigma y + \rho q = \tau c_l. \tag{9}$$

Proof. Write $Q = qc_z + c_l u$, $S = sc_z + c_m d$, $K = kc_z + c_t f$, with $u, d, f \in H_z$. Projection at l, m, z gives $\overline{L} c_t f = 0$, so $f = 0$; projection at l, t, z similarly gives $d = 0$. We claim that

$$Lk + Ps = \tau c_l c_m c_t \tag{10}$$

is impossible when both double projections are nonzero. Its full quotient shows that if one of $\overline{k}, \overline{s}$ is zero, then both are zero. Writing $k = \kappa c_t, s = \sigma c_m$, projection at l, m and l, t forces $\kappa = \sigma = 0$, a contradiction. If both quotients are nonzero, quotient the whole spaces at m, t by $\mathbb{C}s, \mathbb{C}k$. The left side vanishes and the pure target survives, again a contradiction. Thus $R \neq 0$. The H_z component of the original equation is $c_l u R = 0$, giving $u = 0$. Cancel c_z to obtain (8); the same obstruction gives $q \neq 0$. Full projection of that equation is $\overline{L}\,\overline{k} + \overline{P}\,\overline{s} = 0$, proving the zero/nonzero dichotomy. Finally the $H_m c_t, c_m H_t, c_m c_t$ sectors are exactly (9). □

Proposition 3.5. *For* $\operatorname{rank} \overline{L} = \operatorname{rank} \overline{P} = 1$*, the following five families are disjoint and exhaustive.*

I. *Choose $\eta \neq 0$, and take L, P free with projected rank one. Set*

$$Q = \eta c_l c_z, \qquad R = (\tau/\eta) c_m c_t, \qquad S = K = 0.$$

II. *Choose $\kappa \neq 0$, $q \in V_l, r \in V_m$ with $\overline{q}, \overline{r} \neq 0$, and take P free with projected rank one. Set*

$$L = (\tau c_l c_m - qr)/\kappa, \quad Q = qc_z, \quad R = rc_t, \quad S = 0, \quad K = \kappa c_t c_z.$$

III. *Choose $\sigma \neq 0$, $q \in V_l, r \in V_t$ with $\overline{q}, \overline{r} \neq 0$, and take L free with projected rank one. Set*

$$P = (\tau c_l c_t - qr)/\sigma, \quad Q = qc_z, \quad R = c_m r, \quad S = \sigma c_m c_z, \quad K = 0.$$

IV. *Choose $\kappa, \sigma \neq 0$, $q \in V_l$ with $\overline{q} \neq 0$, nonzero $A \in H_m, B \in H_t$, arbitrary $x \in V_l$ and $\rho \in \mathbb{C}$. Put*

$$y = (\tau c_l - \kappa x - \rho q)/\sigma$$

and set

$$\begin{gathered} L = qA + xc_m, \qquad P = qB + yc_t, \qquad Q = qc_z, \\ R = -\kappa A c_t - \sigma c_m B + \rho c_m c_t, \qquad S = \sigma c_m c_z, \qquad K = \kappa c_t c_z. \end{gathered} \tag{11}$$

V. *Choose $\eta \neq 0$, vectors $v \in V_l, s \in V_m, k \in V_t$ with all three quotients nonzero, and arbitrary $\alpha, \beta \in \mathbb{C}$. Set*

$$\begin{gathered} L = vs + \alpha c_l c_m, \qquad P = -vk + \beta c_l c_t, \qquad Q = \eta c_l c_z, \\ R = (\tau c_m c_t - \alpha c_m k - \beta s c_t)/\eta, \qquad S = sc_z, \qquad K = kc_z. \end{gathered} \tag{12}$$

In IV and V all six whole tensors are nonzero and the two projected l-endpoint lines coincide. Independent l-endpoint lines occur only in I–III, with the indicated condition imposed on the free tensors.

Proof. Use Lemma 3.4. If $\overline{k} = \overline{s} = 0$, consider (9). For $\kappa = \sigma = 0$, its first two equations give $r_H = t_H = 0$, and $\rho q = \tau c_l$ gives I. For $\kappa \neq 0, \sigma = 0$, we have $t_H = 0$ and $\kappa L = \tau c_l c_m - q(r_H + \rho c_m)$. The nonzero projected rank forces $\overline{q}, r_H \neq 0$, giving II. The opposite scalar choice gives III. If both scalars are nonzero, set $A = -r_H/\kappa$, $B = -t_H/\sigma$. Then the first two equations give $L_H = qA, P_H = qB$, with $\overline{q}, A, B \neq 0$, and the last equation determines y. This is IV.

Suppose now $\overline{s}, \overline{k} \neq 0$. Quotient (8) at m, t by the actual lines $\mathbb{C}s, \mathbb{C}k$. The target survives, so the remaining equality $qR' = \tau c_l[c_m][c_t]$ forces $q = \eta c_l$ with $\eta \neq 0$. Projection at l gives $L_l k + P_l s = 0$. Quotienting at m by $\mathbb{C}s$ and then canceling the nonzero vectors shows $L_l = us, P_l = -uk$, for a nonzero $u \in H_l$. Consequently $L = us + c_l A_m$ and $P = -uk + c_l B_t$. After substitution, the remaining equation is

$$A_m k + sB_t + \eta R = \tau c_m c_t.$$

Its double quotient is $\overline{A_m}\,\overline{k} + \overline{s}\,\overline{B_t} = 0$. Because $\overline{s}, \overline{k} \neq 0$, contraction by a functional nonzero on either factor gives

$$A_m = \gamma s + \alpha c_m, \qquad B_t = -\gamma k + \beta c_t$$

for scalars γ, α, β. Set $v = u + \gamma c_l$. This gives exactly V.

For completeness the converses are whole-tensor identities. In I only qR contributes. In II and III its non-target part cancels the corresponding part of Lk or Ps. In IV the sum is

$$(\kappa x + \sigma y + \rho q) c_m c_t = \tau c_l c_m c_t,$$

since the terms in qAc_t and $qc_m B$ cancel separately. In V the terms vsk, $\alpha c_l c_m k$, and $\beta c_l s c_t$ each cancel, leaving the same target. Multiplying by c_z proves (3). The prescribed projected ranks follow from the displayed nonzero outer factors.

I–III are distinguished by which of S, K vanish. In IV both are nonzero with zero leaf quotients; in V both leaf quotients are nonzero. Thus the families are disjoint. IV and V visibly have common projected l-endpoint lines. Conversely, if the lines of $\overline{L}$ and $\overline{P}$ are independent, the equation $\overline{L}\,\overline{k} + \overline{P}\,\overline{s} = 0$ forces both leaf quotients zero, and IV still has coincident lines. This gives the last assertion. □

Remark 3.6. With the fixed complements, the parameters in each wedge family are recoverable from the whole tuple. First recover q, s, k by contraction against the covector taking c_z to one and vanishing on H_z. In IV, the unique decomposition of R gives A, B, ρ, and contraction of L at c_m gives x. In V, contraction of L against any covector vanishing on c_m and taking s to one recovers v; the residual pure tensors recover α, β. The simpler recoveries in I–III follow directly from their formulas.

Proposition 3.7. *For* $\operatorname{rank}\overline{L} = 2$, $\operatorname{rank}\overline{P} = 1$, *and the other four projections zero,* $K = 0$ *and* L *is free with projected rank two. The remaining tensors have precisely one of the following forms:*

1. $S = 0$, $Q = ac_l c_z$, $R = (\tau/a) c_m c_t$, *with* $a \neq 0$, *and* P *free with projected rank one;*
2. *for* $\sigma \neq 0$, $u \in V_l, v \in V_t$ *with* $\overline{u}, \overline{v} \neq 0$,

$$S = \sigma c_m c_z, \quad Q = u c_z, \quad R = c_m v, \quad P = (\tau c_l c_t - uv)/\sigma.$$

Proof. In (8), write $\overline{P} = xy$ with $x, y \neq 0$. Full projection gives $\overline{L}\,\overline{k} + x\overline{s}\,y = 0$. A nonzero functional at l annihilating x cannot annihilate the rank-two tensor $\overline{L}$; hence $\overline{k} = 0$, and then $\overline{s} = 0$. In (9), projection of the first equation at l is a rank-two tensor multiplied by κ plus one outer product. Thus $\kappa = 0$, and then $r_H = 0$ because $q \neq 0$. If $\sigma = 0$, the remaining equations give the first branch. Otherwise write $u = q$, $v = t_H + \rho c_t$; they give the second branch and its nonzero quotient conditions. Its whole P has rank two, since (c_l, u) and (c_t, v) are independent pairs. Direct substitution proves both converses, with no further restriction on L. □

Proposition 3.8. *For* $\operatorname{rank}\overline{L} = \operatorname{rank}\overline{P} = 2$*, and the other four projections zero, all solutions are*

$$S = K = 0, \qquad Q = ac_l c_z, \qquad R = (\tau/a)c_m c_t, \qquad a \neq 0,$$

with L, P *free with projected rank two.*

Proof. Full projection of (8) gives $\overline{L}\,\overline{k} + \overline{P}\,\overline{s} = 0$. If $\overline{k} \neq 0$, contract at t with a nonzero functional annihilating it. Rank two of $\overline{P}$ forces $\overline{s} = 0$, and the original equality then forces $\overline{k} = 0$, a contradiction. Thus both leaf quotients vanish. Projecting the first two equations of (9) at l, rank two versus rank at most one gives $\kappa = \sigma = 0$. The remaining equations give the stated pure Q, R. Their product is the target, proving the converse for arbitrary whole lifts of L, P. □

3.4 The two triangle types

The triangle is on l, m, t, with z its isolated projected vertex. The unique pure spoke from Lemma 2.2 distinguishes one of these three vertices. In the next statement call that vertex t. Choosing each of the three possible vertices in turn covers all labelled triangle solutions.

Proposition 3.9. *Choose* $\kappa \neq 0$*,* $q, v \in V_l$*,* $s, w \in V_m$*, and* $h \in H_t$*, with* $\overline{q}, \overline{s}, h \neq 0$*, and set*

$$\begin{aligned} Q = qc_z, \qquad S = sc_z, \qquad K = \kappa c_t c_z, \\ P = qh + vc_t, \qquad R = -sh + wc_t, \qquad L = (\tau c_l c_m - vs - qw)/\kappa. \end{aligned} \tag{13}$$

Extend $\overline{q}$ *to a quotient basis* $(\overline{q}, e)$ *at* l*, and* $\overline{s}$ *to a quotient basis* $(\overline{s}, f)$ *at* m*. Write*

$$\overline{v} = v_1\overline{q} + v_2 e, \qquad \overline{w} = w_1\overline{s} + w_2 f, \qquad d = v_1 + w_1.$$

The exact additional restrictions for the two triangle types are

$$\begin{aligned} &(1,1,1): \quad v_2 w_2 = 0, \quad (d, v_2, w_2) \neq (0,0,0), \\ &(2,1,1): \quad v_2 w_2 \neq 0. \end{aligned} \tag{14}$$

In the second row L *is the rank-two projected edge. These formulas exhaust both types and give their full converses. For the fixed complement* H_t*, their parameters are unique once the pure-spoke vertex and the ordering of the other two vertices are fixed.*

Proof. Apply Lemma 2.2, with its pure-spoke vertex taken as t. It gives the whole forms (13) with a lift of a nonzero quotient vector in the t slot. Changing that lift by $h \mapsto h + \lambda c_t$ changes the remaining vectors by

$$v \mapsto v - \lambda q, \qquad w \mapsto w + \lambda s.$$

These simultaneous changes leave every tensor unchanged, since $vs + qw$ is unchanged. They allow and uniquely determine the choice $h \in H_t$.

The two incident projections are $\overline{P} = \overline{q}\,h$ and $\overline{R} = -\overline{s}\,h$, both nonzero of rank one. The unscaled numerator of the opposite double projection has matrix

$$\overline{v}\,\overline{s} + \overline{q}\,\overline{w} \quad \longleftrightarrow \quad \begin{pmatrix} d & w_2 \\ v_2 & 0 \end{pmatrix}.$$

Its determinant is $-v_2 w_2$. A nonzero matrix of this size has rank one exactly when its determinant vanishes, and rank two exactly when it does not. This proves (14), including the branches $v_2 = 0$ and $w_2 = 0$ and their intersection; the excluded zero triple is exactly a vanished opposite projection.

Conversely the three products from (13) are

$$\begin{aligned} LK &= \tau c_l c_m c_t c_z - v s c_t c_z - q w c_t c_z, \\ PS &= q s h c_z + v s c_t c_z, \\ QR &= -q s h c_z + q w c_t c_z. \end{aligned}$$

Their sum is the required whole target, with all scalar components retained. The rank test gives the desired support exactly. Finally Q, S, K recover q, s, κ by contraction at c_z. The partial t-projection of P recovers the quotient of h, and the fixed complement recovers h itself. The residual tensors in P, R then recover v, w. This proves uniqueness. □

Remark 3.10. All six tensors and all three products are nonzero in either triangle type. For type $(2,1,1)$, the pairs $(\overline{q}, \overline{v})$ and $(\overline{s}, \overline{w})$ are independent. In the bases (c_l, q, v), (c_m, s, w), the matrix of L is

$$\kappa^{-1} \begin{pmatrix} \tau & 0 & 0 \\ 0 & 0 & -1 \\ 0 & -1 & 0 \end{pmatrix}.$$

Thus its whole rank is three, while P, R have whole rank two and the spokes have rank one. For type $(1,1,1)$, either $\operatorname{span}(c_l, q, v)$ or $\operatorname{span}(c_m, s, w)$ has dimension at most two, so all three triangle tensors have rank at most two. Vanishing individual coordinates or further whole-rank drops require no additional chart.

3.5 The four-cycle

Proposition 3.11. *Suppose $\overline{L} = \overline{K} = 0$, and $\overline{P}, \overline{Q}, \overline{R}, \overline{S}$ are nonzero of rank one. All solutions are exactly*

$$\begin{aligned} L = \gamma c_l c_m, \qquad K = (\tau/\gamma) c_t c_z, \qquad \gamma \neq 0, \\ P = u_l u_t, \qquad Q = u_l u_z, \qquad R = u_m u_t, \qquad S = -u_m u_z, \end{aligned} \tag{15}$$

where $u_i \in V_i$ and $\overline{u_i} \neq 0$ for all four sites. All six whole tensors have rank one, and $PS + QR = 0$, $LK = \tau c_l c_m c_t c_z$.

Proof. Full projection gives $\overline{P}\,\overline{S} = -\overline{Q}\,\overline{R} \neq 0$. Equality of nonzero simple four-tensors gives equal endpoint lines at every site. The scalar factors can be chosen consistently: first factor $\overline{P} = a_l a_t$, then $\overline{Q} = a_l a_z$, then $\overline{R} = a_m a_t$; the equality forces $\overline{S} = -a_m a_z$. Here each a_i is a nonzero vector in H_i. Retain all whole lifts by writing

$$\begin{aligned} P &= a_l a_t + p_l c_t + c_l p_t + p_0 c_l c_t, \\ Q &= a_l a_z + q_l c_z + c_l q_z + q_0 c_l c_z, \\ R &= a_m a_t + r_m c_t + c_m r_t + r_0 c_m c_t, \\ S &= -a_m a_z + s_m c_z + c_m s_z + s_0 c_m c_z, \end{aligned}$$

with the subscripted vector at site i in H_i. Since $\overline{L} = \overline{K} = 0$, their product contributes no three-H component. The four three-H components of the full equation therefore give

$$a_l s_m + q_l a_m = 0, \quad -p_l a_m + a_l r_m = 0, \quad a_t s_z + r_t a_z = 0, \quad -p_t a_z + a_t q_z = 0.$$

For nonzero a, b, an equality $ax + yb = 0$ forces $x = \lambda b, y = -\lambda a$, including $\lambda = 0$: quotienting by either fixed line and then canceling the other factor proves this. Consequently there are scalars $\alpha, \beta, \delta, \epsilon$ with

$$\begin{aligned} p_l = \alpha a_l, \quad r_m = \alpha a_m, \quad q_l = -\beta a_l, \quad s_m = \beta a_m, \\ p_t = \delta a_t, \quad q_z = \delta a_z, \quad r_t = \epsilon a_t, \quad s_z = -\epsilon a_z. \end{aligned}$$

Define whole vectors

$$x_l = a_l + \delta c_l, \quad x_m = a_m + \epsilon c_m, \quad x_t = a_t + \alpha c_t, \quad x_z = a_z - \beta c_z.$$

Expansion gives

$$\begin{aligned} P &= x_l x_t + A c_l c_t, & Q &= x_l x_z + B c_l c_z, \\ R &= x_m x_t + C c_m c_t, & S &= -x_m x_z + D c_m c_z, \end{aligned} \tag{16}$$

where $A = p_0 - \delta\alpha$, $B = q_0 + \delta\beta$, $C = r_0 - \epsilon\alpha$, $D = s_0 - \epsilon\beta$. All $\overline{x_i}$ are nonzero.

Choose complements to $\mathbb{C}c_i$ containing x_i; this is merely a choice of basis for the following coefficient comparison, and does not change the tensors or τ. Write the still unrestricted missing edges in these complements as

$$L = \xi_l c_m + c_l \eta_m + \ell c_l c_m, \qquad K = \rho_t c_z + c_t \sigma_z + \kappa c_t c_z.$$

The four cross-pair components, the four single-site components, and the constant component of (3) are respectively

$$\begin{aligned} \xi_l \rho_t &= -D x_l x_t, & \xi_l \sigma_z &= -C x_l x_z, \\ \eta_m \rho_t &= -B x_m x_t, & \eta_m \sigma_z &= A x_m x_z, \\ \kappa\xi_l = \kappa\eta_m = \ell\rho_t = \ell\sigma_z &= 0, & \ell\kappa + AD + BC &= \tau. \end{aligned} \tag{17}$$

These are all remaining coefficients: the other two pair components are identically zero, the three-site components have just been eliminated, and the four-site components cancel in (16).

Multiply the first and fourth pair equations in (17), and then the second and third, restoring both to the same site order. Their left sides are the identical tensor $\xi_l \eta_m \rho_t \sigma_z$. Since $x_l x_m x_t x_z \neq 0$, their right sides give $-DA = CB$, so $AD + BC = 0$. This argument divides by none of the unknown vectors or residual scalars. The final equation now gives $\ell\kappa = \tau \neq 0$. The single-site equations force $\xi_l = \eta_m = \rho_t = \sigma_z = 0$, and the pair equations force $A = B = C = D = 0$. Taking $\gamma = \ell$ and $u_i = x_i$ yields (15) in the original spaces.

Conversely let the parameters in (15) be given, and put $U = u_l u_m u_t u_z$. Then $LK = \tau c_l c_m c_t c_z$, $PS = -U$, and $QR = U$. The double projections have exactly the prescribed ranks, and every whole factor is nonzero of rank one. This proves the full converse, including arbitrary c-components in the four vectors. □

The propositions describe whole tensors, rather than only quotient data. Every displayed parametrization therefore includes all permitted cancellations and whole-rank drops. The list is finite at the level of support and rank types; Proposition 3.1 uses geometric parameters, and no classification into finitely many orbits is asserted.

4 Dimensions of the strata

We use algebraic dimension for constructible strata, equivalently the dimension of their closures. A whole block with zero double projection belongs to a five-dimensional vector space. The loci of blocks with projected ranks one and two have dimensions 8 and 9, respectively: the corresponding loci in $\mathrm{Mat}_{2\times 2}(\mathbb{C})$ have dimensions 3 and 4, and the five target-containing entries are free.

4.1 The empty projected support

The empty-support chart identifies a point with a vector $a \in E = \mathbb{C}^6$ and four maps $B_i : H_i^* \to S_i$, where S_i is the coordinate star at i, satisfying

$$q(a) = \tau, \qquad U = \sum_i \operatorname{im} B_i \subset a^\perp, \qquad q|_U = 0, \qquad q(x) = x_{12}x_{34} + x_{13}x_{24} + x_{14}x_{23}. \tag{18}$$

Here perpendicularity uses the polar form of q. The subspace $a^\perp$ is nondegenerate of dimension five, so $r = \dim U \leq 2$.

Proposition 4.1. *The stratum with all six double projections zero has dimension* 11.

Proof. The actual maps determine U. On fixed-rank Grassmannian charts, recovery of their image is regular. We may therefore count the incidence data $(U, a, (B_i))$, imposing the open condition that the images span U, without adding an artificial subspace parameter.

Identify E with $\bigwedge^2 \mathbb{C}^4$, changing the sign of the 13 coordinate. Then q becomes the Pfaffian, and $S_i = e_i \wedge \mathbb{C}^4$. A nonzero isotropic bivector is decomposable. An isotropic two-plane U is consequently of the form

$$U = \ell \wedge H, \qquad \ell \subset H \subset \mathbb{C}^4, \quad \dim \ell = 1, \quad \dim H = 3,$$

for a unique flag. Indeed two independent decomposable generators have wedge zero, so their supporting two-planes meet in a line and span a three-space. This also proves the converse.

First suppose $r = 2$ and $\ell = \mathbb{C}e_i$. Then $U \subset S_i$. It contains $m = 0, 1$, or 2 of the three coordinate axes of S_i, and these alternatives have parameter dimensions $2 - m$. For $j \neq i$, $U \cap S_j$ is a line precisely when the corresponding coordinate edge lies in U. Hence $\sum_j \dim(U \cap S_j) = 2+m$. For any fixed isotropic two-plane, $U^\perp$ has dimension four, radical U, and a nondegenerate two-dimensional quotient. The set $q(a) = \tau$ in $U^\perp$ is thus nonempty of dimension three. The maps B_j contribute $2(2 + m)$ parameters. The total dimension is

$$(2 - m) + 3 + 2(2 + m) = 9 + m \leq 11.$$

The spanning condition is a nonempty open condition.

If ℓ is not a coordinate line, then U is contained in no star, and $U \cap S_i$ has dimension one exactly when $e_i \in H$. Let k be the number of coordinate axes contained in H. Spanning U by these intersections requires $k \geq 2$, and $k \leq 3$. For $k = 2$, H varies in an open subset of $\mathbf{P}^1$; ℓ varies in $\mathbf{P}(H)$ outside the plane spanned by those two axes, so the flag parameter dimension is three. The maps contribute four and a contributes three, for a total of ten. For $k = 3$, H is a coordinate three-space and ℓ varies in its projective plane, excluding the coordinate points. The flag dimension is two, the maps contribute six, and a contributes three, for a total of eleven. Here the three nonzero intersections are the lines $\ell \wedge e_i$ for the three axes $\mathbb{C}e_i \subset H$; wedging with ℓ identifies H/ℓ with U, and the images of these axes span H/ℓ, so the three lines span U. If ℓ lies in the plane of two of these axes, their two lines coincide and the third differs; each $U \cap S_i$ is still a line, so the maps still contribute six and spanning is still a nonempty open condition. These are all the two-plane cases.

If $r = 1$, U must lie in at least one star. A non-coordinate line lies in exactly one star and has two line parameters and two map parameters. A coordinate edge line has no line parameter and four map parameters, from its two incident stars. For either type of isotropic line, $q(a) = \tau$ in $U^\perp$ has dimension four. Thus $r = 1$ contributes dimension eight. For $r = 0$ the dimension is five.

The incidence alternatives are finite and retain all intersection jumps. They give an upper bound of eleven. Equality is also visible in (1) by requiring both free blocks to have zero double projection: this is $\mathbb{A}^{10} \times \mathbb{G}_m$, of dimension eleven. □

4.2 The remaining charts

The following counts use fixed complements H_i and whole-vector parameters. They do not count arbitrary changes of basis as extra parameters. In every chart whose dimension is asserted, the parameters are recoverable regularly on finitely many nonzero-coordinate charts. This recovery is given for the wedges of ranks $(1,1)$ in Remark 3.6 and for the triangles in Proposition 3.9; for the single-edge charts, the other wedges and the rows of the crossing table it is written below, where each is counted; for the four-cycle, the end of this subsection identifies the redundancy exactly.

For a single rank-one projected edge L, write $K = \kappa c_t c_z$. When $\kappa \neq 0$ and both crossing products are nonzero, the four vectors p, q, r, s contribute twelve parameters and κ contributes one. The condition

$$\operatorname{rank}\big(\bar{p}\bar{s} + \bar{q}\bar{r}\big) = 1$$

cuts a genuine hypersurface: the determinant is, up to sign, the product of the two determinants of the endpoint vector pairs. The nonzero-matrix condition removes neither entire hypersurface component. This chart has dimension twelve. With one crossing product zero, there are κ, two whole vectors, and a free zero-projection partner, again giving $1+3+3+5 = 12$. In these $\kappa \neq 0$ charts K gives κ as its $c_t c_z$ coefficient, contraction at c_t or c_z recovers each named vector from its block among P, Q, R, S, and a free partner is itself a block. When $\kappa = 0$, the free rank-one-projected L contributes eight. A zero crossing product leaves the scalar of the two pure blocks whose product is the target, and a free zero-projection block, giving $8+1+5 = 14$. If both products are nonzero, each exceptional-site chart has two nonzero scalars and one free whole vector, with the other vector solved by the target equation; its dimension is $8+5 = 13$. For $\kappa = 0$, L and any free block are blocks of the tuple, and the scalar of a zero crossing product is the coefficient of the pure block Q in the first case of Lemma 3.2 and of P in the second. In each row of the crossing table the two scalars are the coefficients of its two pure blocks, and contraction at the c factor shown there recovers u from P or S. The single rank-one edge stratum therefore has dimension fourteen.

For a single rank-two projected edge, the $\kappa \neq 0$ chart has dimension thirteen, since its endpoint independence conditions are open. For $\kappa = 0$, the free block has dimension nine, so the two preceding crossing alternatives give dimensions fifteen and fourteen. This stratum has dimension fifteen. In its charts the parameters are recovered as for rank one.

For a rank-$(1,1)$ wedge, the five families in the atlas have dimensions

$$17, \quad 15, \quad 15, \quad 13, \quad 12, \tag{19}$$

respectively. In the first family these are two free blocks of dimension eight and one nonzero scalar. In the second and third they are a free block of dimension eight, two whole vectors and a nonzero scalar. In the fourth, the free parameters are two scalars, a whole vector q, two vectors $A \in H_m, B \in H_t$, a whole vector x, and one scalar ρ; their dimensions sum to $2+3+2+2+3+1 = 13$, and y is determined. In the fifth they are $\eta, v, s, k, \alpha, \beta$, of total dimension $1+3+3+3+1+1 = 12$. All stated nonvanishing conditions are open, and the fixed-coordinate recovery in the atlas removes any ambiguity.

A rank-$(2,1)$ wedge has two charts. Their dimensions are $9+8+1 = 18$ and $9+1+3+3 = 16$. A rank-$(2,2)$ wedge has dimension $9+9+1 = 19$. These counts include the arbitrary whole lifts of its free blocks. In these charts the free blocks are blocks of the tuple. The scalar is the $c_l c_z$ coefficient of Q, except in the second $(2,1)$ chart, where it is the $c_m c_z$ coefficient of S and contraction of Q at c_z and of R at c_m recovers the two vectors.

For triangles use, for example, the pure spoke $K = \kappa c_t c_z$. The common chart is

$$\begin{gathered} Q = q c_z, \quad S = s c_z, \quad K = \kappa c_t c_z, \\ P = qh + v c_t, \quad R = -sh + w c_t, \quad L = \kappa^{-1}(\tau c_l c_m - vs - qw). \end{gathered} \tag{20}$$

Require $h \in H_t \setminus \{0\}$; this removes the entire additive ambiguity in choosing its whole lift. The parameters q, s, v, w, h, κ have total dimension fifteen. For triangle ranks 211, the two quotient endpoint pairs $(\bar{q}, \bar{v})$ and $(\bar{s}, \bar{w})$ are independent; these are open nonempty conditions. For ranks 111, the opposite projected matrix is nonzero and its determinant vanishes. Its determinant factors as the product of those two endpoint determinants, giving a union of hypersurfaces in this parameter domain. Its dimension is fourteen. There are only three possible pure-spoke branches for a fixed triangle.

Finally the four-cycle chart has a nonzero scalar and four whole vectors outside their target lines, giving thirteen parameters. The entire redundancy is

$$(u_l, u_m, u_t, u_z) \longmapsto (au_l, au_m, a^{-1}u_t, a^{-1}u_z), \qquad a \in \mathbb{G}_m.$$

Equality of the three products $u_l u_t, u_l u_z, u_m u_t$ forces exactly this relation; the fourth then agrees. Normalizing a nonzero quotient coordinate of u_l gives finite parameter charts of dimension twelve.

Together with Proposition 4.1, these calculations prove the full-stratum dimension column of Table 1.

5 Maximal components and nonzero summands

Proof of Theorem 1.1(1). For each centred wedge let $X_{i;jk}$ be the set (1). It is closed in F_τ: impose the two zero-block conditions and the two pure-line conditions. The fibre equation then requires the two pure scalar coefficients to satisfy $qr = \tau$. Consequently

$$X_{i;jk} \simeq \mathbb{A}^{18} \times \operatorname{Spec} \mathbb{C}[q, r]/(qr - \tau) \simeq \mathbb{A}^{18} \times \mathbb{G}_m.$$

Its rank-$(2, 2)$ projected subset is nonempty and dense, so its closure is exactly $X_{i;jk}$. There are $4\binom{3}{2} = 12$ centred wedges. Their varieties are distinct because their dense projected supports differ.

The fibre is a finite union of its labelled strata. All strata other than rank-$(2, 2)$ wedges have dimension at most eighteen. Taking closures gives a finite closed cover by the twelve $X_{i;jk}$ and sets of dimension at most eighteen. Hence $\dim F_\tau = 19$. An irreducible component of dimension nineteen must be contained in one of these twelve closed irreducible sets, and maximality makes it equal to that set. Conversely each $X_{i;jk}$ is such a component, since any component containing it has dimension nineteen and must be one of the twelve. This proves both assertions. □

Proof of Theorem 1.1(2). Intersect every chart with the condition that all six whole blocks are nonzero. For a single projected edge only the $\kappa \neq 0$ chart with both crossing products nonzero survives, giving dimensions twelve and thirteen for projected ranks one and two. For a rank-$(1, 1)$ wedge only its fourth and fifth families survive, of dimensions thirteen and twelve. Wedges of ranks 21 and 22 force a whole block to vanish. Both triangle types and the four-cycle are entirely nonzero-block charts. The empty support has dimension at most eleven. This proves the last column of Table 1 and gives $\dim F_\tau^{\mathrm{nz}} = 15$.

The dimension-fifteen strata are precisely the rank-$(2, 1, 1)$ triangles. There are twelve labelled choices: an isolated projected vertex and a high edge among the other three vertices. For any fixed choice, the chart (20), with the fixed-complement convention, has parameter domain

$$\kappa \in \mathbb{G}_m, \quad h \in H_t \setminus \{0\}, \quad (\bar{q}, \bar{v}), (\bar{s}, \bar{w}) \text{ independent.}$$

It is a nonempty open subset of affine fifteen-space. The recovery of its actual parameters is regular on nonzero-coordinate charts. Hence this stratum is irreducible of dimension fifteen.

Its closure within F_τ^{nz} is therefore irreducible of dimension fifteen. These twelve closures are distinct. Different isolated vertices impose different closed zero-projection equations. For a fixed isolated vertex, different high edges are separated by the closed rank-at-most-one conditions on the two low edges. Each other triangle stratum violates one of the relevant closed conditions. All remaining active strata have dimension at most fourteen. The same finite-closed-cover argument as above now proves that the twelve closures are exactly the components of maximal dimension in F_τ^{nz}. □

Proof of Theorem 1.1(3). For nonzero vector spaces U, V, the factorization map

$$(U \setminus 0) \times (V \setminus 0) \longrightarrow \{a \otimes b : a, b \neq 0\}$$

has fibre $\mathbb{G}_m$, acting by $(a, b) \mapsto (sa, s^{-1}b)$. It is Zariski locally trivial. In chosen bases, on the chart $T_{pq} \neq 0$ one normalized factorization is

$$a_p = 1, \qquad b_j = T_{pj}, \qquad a_i = T_{iq}/T_{pq}.$$

The rank-one equations imply $T_{ij} = a_i b_j$.

Apply this independently to each of the three paired partitions and restrict to $A + B + C = \tau c_1 c_2 c_3 c_4$. The result is a surjective locally trivial $(\mathbb{G}_m)^3$-bundle $F_\tau^{\mathrm{nz}} \to D_\tau$. Dimensions therefore differ by three. The bundle also induces a bijection on irreducible components. Indeed its inverse image over an irreducible base component is locally a product with an irreducible torus; a dense trivializing open set has irreducible inverse image, whose closure fills that inverse image. For the last claim, a point of that inverse image lies over a second trivializing open set; the two sets meet the irreducible base component in a common dense open subset, and in the product chart over the second set the inverse image of that subset is dense, so the point lies in the closure. Conversely maximal irreducible closed subsets upstairs project densely onto base components, by the local product description. Thus all component dimensions decrease by three, proving the claim. □

Remark 5.1. The twelve largest components of F_τ are disjoint from F_τ^{nz}. The active components of dimension fifteen therefore come from lower-dimensional components of the full fibre. We do not claim that every active point belongs to one of the twelve largest active components, nor identify their boundary intersections.

6 Examples and scope

For a rank-$(2, 1, 1)$ triangle, choose bases (c_i, a_i, b_i) and set

$$\begin{gathered} Q = a_l c_z, \quad S = a_m c_z, \quad K = c_t c_z, \\ P = a_l a_t + b_l c_t, \qquad R = -a_m a_t + b_m c_t, \\ L = \tau c_l c_m - b_l a_m - a_l b_m. \end{gathered}$$

Direct substitution gives $LK + PS + QR = \tau c_l c_m c_t c_z$. The whole ranks of L, P, Q, R, S, K are respectively $3, 2, 1, 2, 1, 1$. The projected triangle has ranks $2, 1, 1$; every matching product is nonzero. The open family (20) explains the fifteen-dimensional stratum containing such points.

The empty projected support can also exhibit nonzero cancellation. Use one quotient variable x at site 1 and one y at site 2, and consider the affine edge forms

$$E_{12} = 1 - 2y, \quad E_{13} = 1 + x, \quad E_{14} = 1 - x, \quad E_{23} = 1 + y, \quad E_{24} = 1 + y, \quad E_{34} = 1.$$

They satisfy

$$E_{12}E_{34} + E_{13}E_{24} + E_{14}E_{23} = 3.$$

Homogenizing separately at each site with its coordinate c_i gives six nonzero whole edge tensors with zero double projections. The corresponding isotropic image U has dimension two. Multiplying all six edges by a scalar s with $3s^2 = \tau$ gives the prescribed target. This example illustrates why projected support alone does not determine whole-block support.

The assumptions $\tau \neq 0$ and $\dim V_i = 3$ are used explicitly. The nonzero target is essential in the contradiction that proves Lemma 2.1 and in the reductions to a pure spoke. The quotient dimension two bounds the projected ranks and enters the incidence count. We make no assertion for the zero fibre, for other local dimensions, or for scheme multiplicities. The lower-dimensional components and the incidence relations between the listed strata remain further questions.

Proof artifacts and AI use

The author, who publishes under the pseudonym Ignoramus, directed this project and takes responsibility for the paper. The author used OpenAI Codex to produce the results, proofs, checking program and text of this paper, and Anthropic Claude to check and revise it as described below, and has not verified the mathematics; no human has checked the proofs. No human peer review or formal proof-assistant verification is claimed, and no AI system is listed as an author. The attribution to OpenAI Codex, of this work and of the Codex checks described below, rests on the project's records and the author's account; which model versions were used at each step has not been verified.

All checks described in this section were made by AI agents. The classifications of the single edge of projected rank two, of the wedge of projected ranks $(2, 1)$ and of the empty projected support, Lemma 2.2 in its present form, the assembly into nine types, and every dimension and component count, Theorem 1.1 included, were first obtained in the review that led to this paper, shortly before it was drafted; the other results go back to earlier stages of the project, some of them in a weaker form. In that review, before drafting, agents of the Codex workflow checked the arguments and compared the results with published work. Two further agents of that workflow then reviewed the manuscript: one reported that it had reconstructed the proofs analytically, claiming no independent machine computation, and that it had found no error or gap requiring repair; the other checked the claims of novelty, the citations and the wording. Both had seen the workflow's earlier assessments of the results, so neither review was blind.

Anthropic Claude prepared this revision at the author's instruction. It wrote this section and the abstract. In Section 1 it wrote the paragraph that defines pure tensors, the definitions of terms, the sentence that names the components of dimension fifteen, the sentence on searches for earlier statements, and the sentences that cite the author's earlier papers and [10]; it added the reference [11], made the pointers to the literature exact and reworded the comparison with the note [5]. It also changed wording in other places and renamed letters that carried two meanings in the proofs of Lemmas 2.1 and 2.2 and in the treatment of the empty projected support, which changed letters in six displayed formulas. It added short sentences that write out steps the earlier text left to the reader: the uniqueness in Lemma 2.2, an instance of each of the nine types, one point each in the proofs of Propositions 3.1 and 4.1 and of Theorem 1.1, and, in Section 4.2, how the parameters are recovered in the single-edge charts, in the rows of the crossing table and in the wedges of ranks $(2, 1)$ and $(2, 2)$. Exact programs, not posted with the paper, tested these sentences on examples; for the step in the proof of Theorem 1.1(3) they tested only the two facts it uses. It

changed no statement of a theorem, proposition, lemma or remark, and it removed or altered no step of the earlier proofs.

Claude agents also checked the paper. One read every line of the unrevised text and found no error and no gap affecting a theorem. One compared the paper with the author's four earlier papers and found none of its results there. A fact-check compared the citations then in the text with the cited works and led to corrections. The Claude session that coordinated this revision compared its own sentences citing the author's earlier papers with those papers, and an earlier draft of this section with the project's records; a further agent audited every verdict, and this section was corrected. Three agents, each working alone, then read the whole text as critics, before the corrections that followed and before the short added sentences described above were written; each checked every numbered statement by hand and tested the parametrizations, the dimensions and the list of nine types with programs of its own. None found a mathematical error; each found faults of wording, credit and definition, most of which were then corrected. The corrections and the added sentences were then compared line by line with the unrevised text by one further Claude agent that did not write them; it checked the added sentences by hand and with programs of its own, and the references added after the fact-check against their sources. Eight of the ten corrections it proposed were then made, in this section and in four sentences elsewhere. Of the other two, the one on the attribution to OpenAI Codex was left to the author, and the other was set aside on the record of an earlier check. None of the checks named in this section is journal peer review, a formal verification or a certificate of originality or priority.

No step of a proof depends on machine computation. The program `check_examples.py`, posted with the paper together with its saved results and a README, checks 20 examples with rational entries, covering all nine projected support and rank types, each under all 24 permutations of the four sites, in exact rational arithmetic without floating point. For each example and permutation it checks all 81 coefficients of $LK + PS + QR = \tau c_l c_m c_t c_z$, the whole and projected ranks of the six blocks, the projected support type, and which blocks and which matching products are zero. For the empty-support examples it also checks, in the notation of Proposition 3.1, that $q(a) = \tau$ and that $\sum_i \operatorname{im} B_i$ is totally isotropic, orthogonal to a and of dimension two. The examples include the two of the preceding section. All but one use $\tau = 3$; one uses $\tau = 12$. These are finite checks of examples. They do not prove that the nine types are exhaustive or that the parametrizations are complete, and they do not check the dimension or component counts.

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